

## Paper 2

### **AI-Assisted Portfolio Triage for Canada's Major Projects: Churn and Net Additions, 2018–2024**

**Running Title:** AI Portfolio Triage – Churn and Additions

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## Abstract

Canada's economic growth and net-zero ambitions rely on converting large project proposals into operating facilities, yet the development portfolio exhibits high volatility and low growth. Existing studies note megaproject delays but lack national, cross-sector metrics, obscuring where attrition occurs. This paper fills that gap by transforming Natural Resources Canada's Major Projects Inventory into reconciled yearly snapshots that reveal high turnover and a chronic graduation bottleneck. We classified 4,060 transitions involving 1,250 projects by sector, size and ownership covering the period of 2017-24, then traced freshmen, graduations and dropouts to calculate churn and net additions. Annual churn averages 230 projects, yet only 45 percent of exits graduate to construction. Results show Alberta and British Columbia both experience heavy churn while their trajectories diverge, Ontario is marginally negative, and Québec, Manitoba and Northwest Territories grow; hydrocarbons and traditional grids retreat while critical-mineral mining, renewable and low-carbon-fuel proposals expand. Only mid-sized ventures valued between \$100 million and \$999 million post net gains; public entities add 26 projects, private firms lose 38. Data-driven surveillance and AI-assisted strategic planning combined with streamlined approvals, targeted public financing, and robust public anchoring offer the most pragmatic path to converting proposals into built assets.

## Introduction

Natural Resources Canada's Major Projects Inventory (MPI) is a rare longitudinal dataset. Each annual release captures about 500 large energy, mining and forestry proposals worth at least \$50 million. Paper 1 treated the MPI as a backward-looking stock taking. What is missing is a dynamic view: how many projects enter and exit each year, which provinces or sectors drive net growth, and where attrition erodes the pipeline. This paper addresses that gap by measuring annual churn and net additions across the 2018–2024 period. We pose a single research question: How volatile is Canada's major-project pipeline, and what factors explain observed growth or contraction? Our contribution is two-fold. First, we convert static MPI tables into reconciled annual snapshots, overcoming data-quality hurdles that previously impeded longitudinal analysis. Second, we provide the first national-level churn metrics, segmented by province, sector, cost bracket, ownership type, and cleantech orientation. By exposing where and why the pipeline expands or withers, the study informs regulators, investors, and other stakeholders tasked with expanding development resources and bandwidth.

## Methods

We analyzed Natural Resources Canada's MPI master file, which reports the status of large Canadian infrastructure projects year by year. The study spans the seven reporting years from 2018 through 2024; each row in the file therefore represents a single project in a single calendar year. Every record was linked across years with a unique project identifier, after which we standardized company and project names, province abbreviations, and sector labels. Constant attributes were added to enrich the data, including company type (public or private), a sub-sector

tag, a broad capital-cost bracket, and a cleantech flag indicating whether the project involves emissions-reducing or renewable technology.

For each calendar year  $t$  we extracted the set of project identifiers and compared it with the set from the previous year,  $t - 1$ . This comparison generated four mutually exclusive outcomes. A project that appeared for the first time was labeled a freshman. If an identifier vanished and its last recorded status was “Under Construction,” the project was classified as a graduate; if it disappeared while still in “Announced & Planning,” “In Review,” or “Approved,” it was treated as a dropout. Projects present in both years were marked as continuing. The opening inventory in any year exactly equaled the previous year’s closing snapshot, confirming the integrity of the roll-forward. We then counted freshmen, graduates, and dropouts for each year and verified that the opening inventory plus net additions equals closing inventory held without exception, where net additions are defined as freshmen minus the combined total of graduates and dropouts. A Boolean flag was set to “true” for every year once this reconciliation passed.

These transition counts were stored in a “transitions” table that records the project identifier, calendar year, transition state, and cumulative years listed in the MPI. The table was merged back to the constant project attributes to enable breakdowns by province, sector, company type, capital bracket, and cleantech status. Two derived metrics emerge from this framework. Project churn captures total annual turnover, the sum of all entries and exits between consecutive year-end snapshots, while net additions measure the net change in pipeline size. Together, churn and net additions reveal whether Canada’s major-project pipeline is expanding, stable, or contracting from one year to the next, and they provide a validated foundation on which more sophisticated analyses, such as survival rate and survival time modelling can be layered.

A note about limitations: This analysis rests on the annual MPI, which only lists proposals above a size threshold drawn from publicly available sources and may lag real-time status changes.

Projects that disappear from the dataset are treated as dropouts even though some could be paused rather than cancelled, while freshman counts may include rebranded legacy plans. The MPI omits key variables, such as permitting milestones, Indigenous and community agreements, and capital-cost escalation as well as macro indicators, such as commodity prices, interest rates, GDP growth, and change in government. Together, these constraints mean observed churn and net additions should be interpreted with caution.

## **Results & Discussion**

### *Pipeline size, churn and net additions*

Canada's major project pipeline has recorded a compounded rate of growth of just 1 percent between 2018 and 2024. Figure 1 and Table 1 show opening inventories hover near 500 projects. Annual churn averages 230 projects, roughly 40 percent of the stock, showing high portfolio volatility. Freshmen inflows generally offset exits. Graduating projects (those that begin construction) run in the 35-80 range each year, while cancellations (dropouts) typically match or exceed graduates except in growth years such as 2022. Net additions swing between -47 and +40 projects, so the closing inventory seldom deviates more than 5 percent from the prior year. In short, the MPI neither sharply expands nor contracts; instead, new proposals continually replace projects that leave, either by graduation or dropping out, keeping the national queue flat.

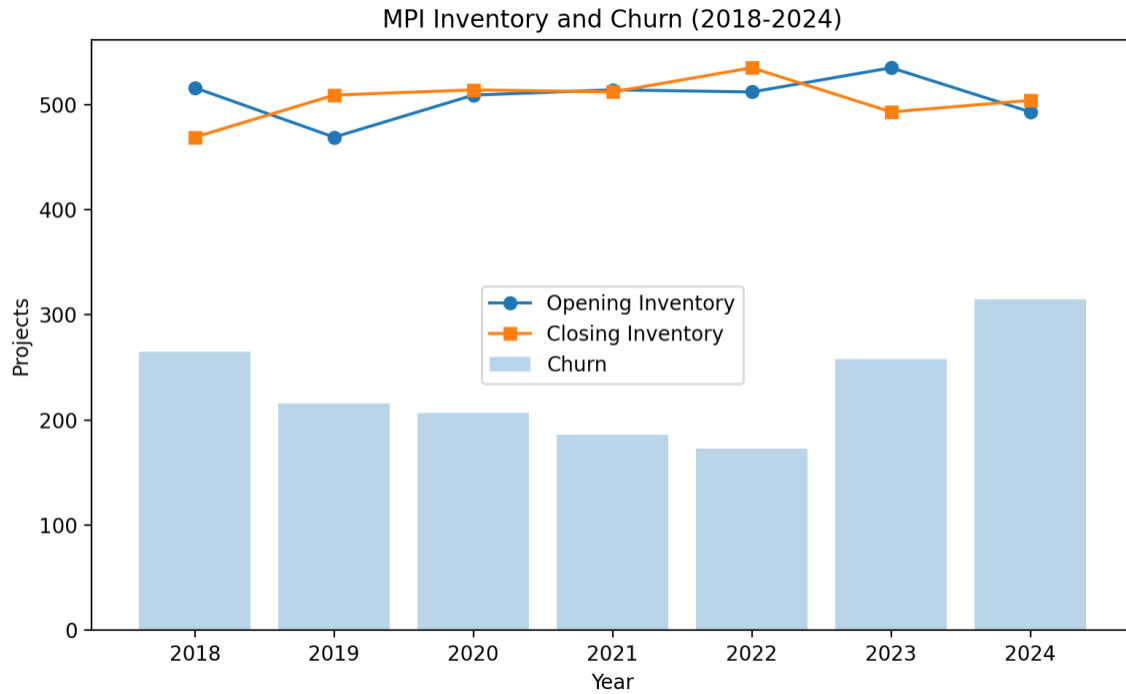


Figure 1: MPI Inventory and Churn, (2018 – 2024).

| Year | Opening | Closing | Freshmen | Graduates | Dropouts | Churn | Net Additions |
|------|---------|---------|----------|-----------|----------|-------|---------------|
| 2018 | 516     | 469     | 109      | 80        | 76       | 265   | -47           |
| 2019 | 469     | 509     | 128      | 41        | 47       | 216   | 40            |
| 2020 | 509     | 514     | 106      | 49        | 52       | 207   | 5             |
| 2021 | 514     | 512     | 92       | 36        | 58       | 186   | -2            |
| 2022 | 512     | 535     | 98       | 33        | 42       | 173   | 23            |
| 2023 | 535     | 493     | 108      | 46        | 104      | 258   | -42           |
| 2024 | 493     | 504     | 163      | 64        | 88       | 315   | 11            |

Table 1: MPI Inventory Turnover and Net Additions, (2018 – 2024).

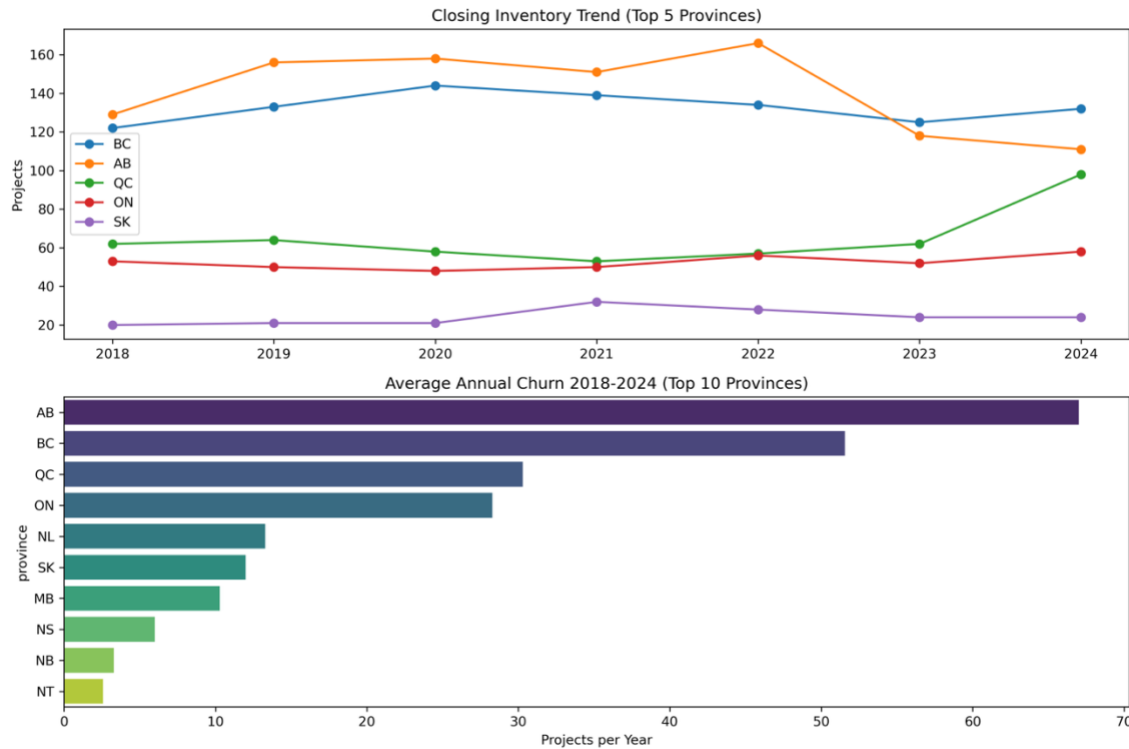
### Provincial churn

Figure 2 highlights that provincial picture is one of steady churn with wide regional contrasts.

Alberta and British Columbia dominate the pipeline by sheer volume. Each finish 2024 with 150-plus projects and shows the highest annual inflows, yet they also record the most exits.

Ontario and Québec display lower and more stable project inventories. Saskatchewan and Newfoundland have the most volatile pipelines, swinging from growth to contraction as

individual megaprojects appear or disappear. Across all provinces, average yearly churn (entries plus exits) ranges from about five projects in the smallest jurisdictions to over 60 in Alberta.



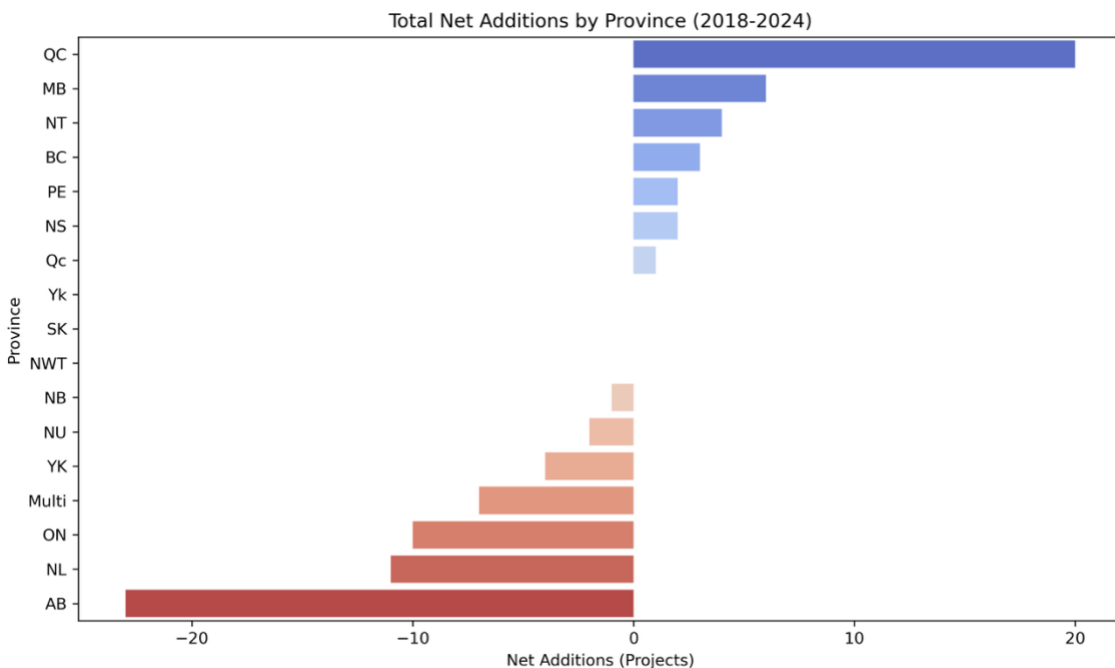
*Figure 2: Closing Inventories and Churn by Year and Province.*

### *Provincial net additions*

Figure 3 illustrates net additions by province. Across the seven-year window, Québec stands out as the clear growth story, expanding its major-project pipeline by a net 20 projects. Although the province experiences brisk turnover, fresh announcements have consistently outnumbered departures. Manitoba and the Northwest Territories also manage to edge their inventories upward, albeit from much smaller bases. In both cases, just a few durable resource or infrastructure proposals tip the balance into positive territory.

The largest contraction occurs in Alberta, whose inventory shrinks by 23 projects. Here heavy churn combines with a recent wave of cancellations in upstream oil and gas, erasing the gains generated by several projects that did advance to construction. Newfoundland and Labrador and Ontario likewise post net declines. In Ontario the queue thins because a steady stream of graduations is not matched by new entries, while Newfoundland and Labrador's drop reflects a lull in large offshore and hydro developments.

Taken together, the figures underline a simple dynamic: a province's net additions depend on whether inflows of brand-new proposals can outrun the combined pull of projects that graduate and those that drop out. Sector composition, policy incentives, and the timing of mega-project cycles all interact to determine whether a regional pipeline deepens or erodes over time.



*Figure 3: Net Additions by Province.*

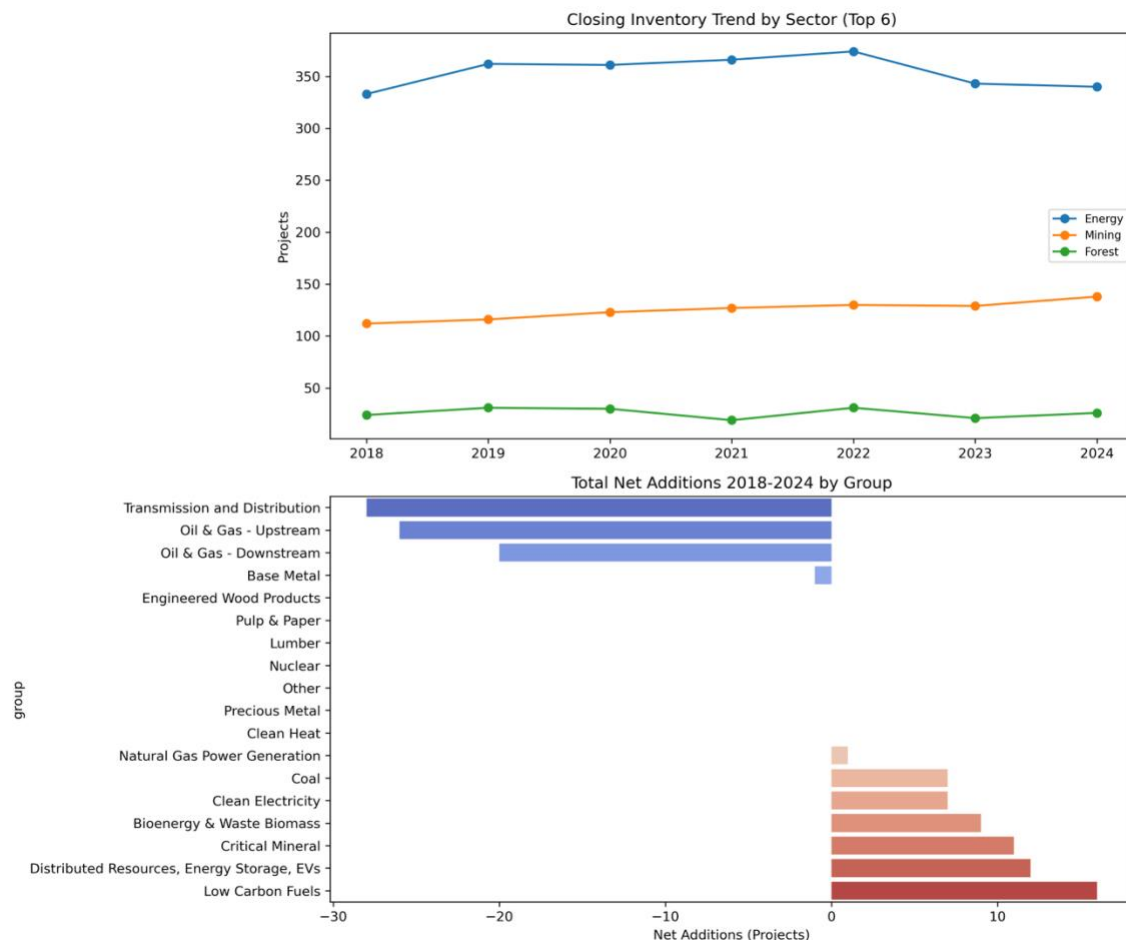
*Sectoral churn*



The MPI pipeline is dominated by a handful of heavyweight sectors and technology groups, but their fortunes diverge sharply, as shown in Figure 4. Energy is the workhorse: it cycles more than 1,100 project entries and exits, half of all churns, yet ends the period with a small net loss (-46). Mining is a distant second on churn but quietly adds 29 projects, buoyed by critical-mineral proposals. Forestry remains small, with single-digit gains.

From a technological lens, clean electricity (wind, solar, hydro) shows the highest turnover with nearly 500 project movements, while still adding seven net projects, suggesting a healthy renewal rate. transmission and distribution and oil and gas groups, by contrast, churn heavily and shrink, reflecting graduations to construction and cancellations outpacing new filings. Low-carbon fuels and distributed-energy segments record modest but positive net growth, hinting at emerging momentum.

Overall, the data reveal a pipeline in transition: traditional hydrocarbon and grid assets are rotating out faster than they are replaced, whereas critical-mineral mining, renewables, and new-energy niches post incremental gains. The sector-group mix therefore tilts gradually toward lower-carbon and electrification themes.



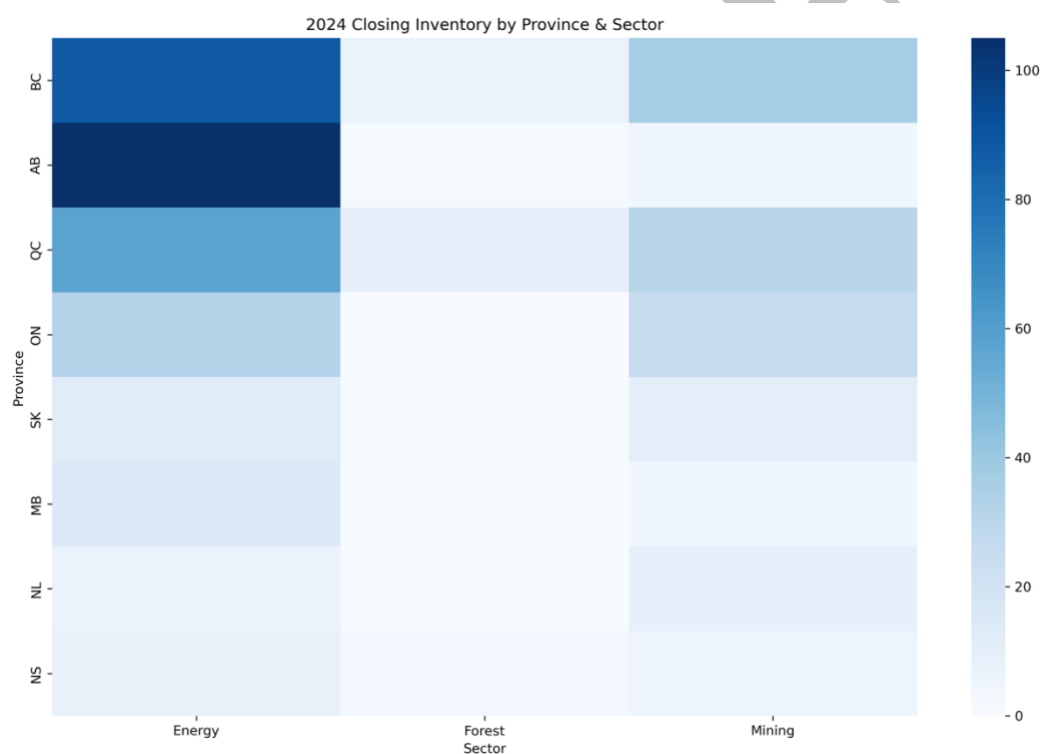
*Figure 4: Closing Inventories and Net Additions by Sector and Group.*

#### *Segmentation by region and sector*

Figure 5 is a project heat-map confirming how sharply the national pipeline is segmented by region and sector. First, energy proposals are ubiquitous yet overwhelmingly concentrated in the West. Alberta and British Columbia each support well over 100 active energy projects, whereas Ontario and Québec report only a few dozen apiece. Even grid-expansion work follows this pattern. Most transmission and distribution proposals cluster west of the Prairies, underpinning the build-out required for LNG export terminals and large-scale renewable power.

Mining follows a different geography. Uranium and critical-mineral developments form a tight north-and-Prairie footprint, with Saskatchewan and the three Territories accounting for the lion's share of new discoveries and processing plants. Québec's mining slate is smaller but expanding as the province positions itself in battery-metal supply chains.

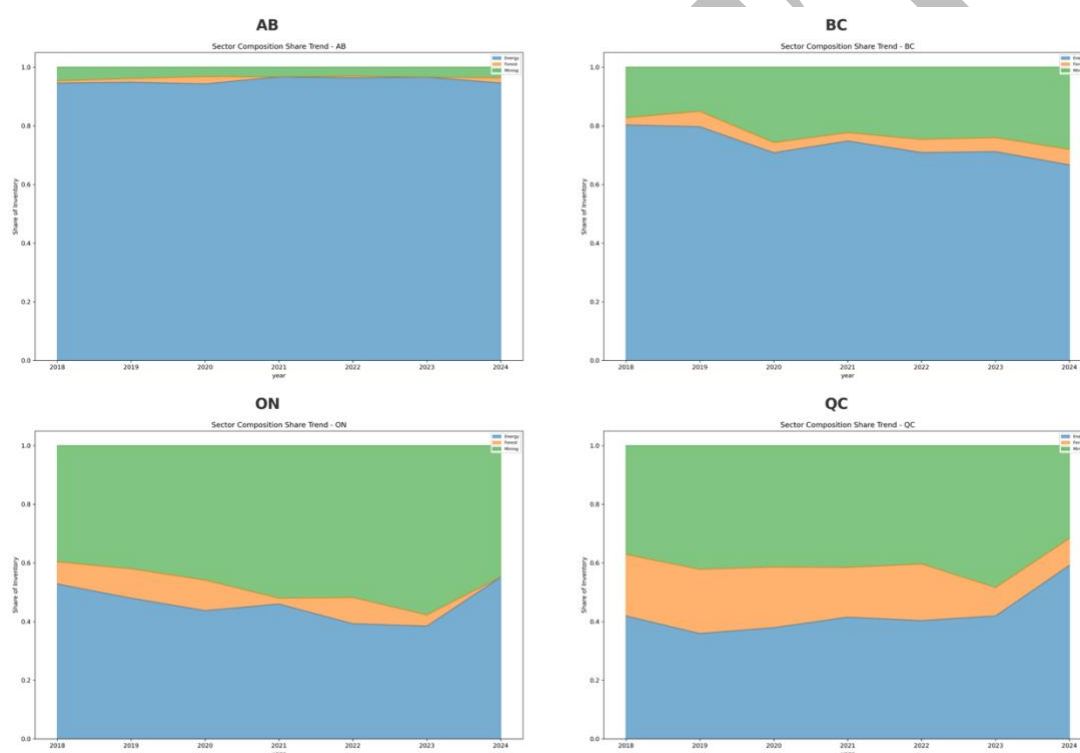
Finally, each province nurtures distinctive niches. Québec retains an outsized forest sector, ranging from bio-refineries to pulp-and-paper modernizations. Newfoundland and Labrador's project list is dominated by hydro and offshore petroleum work. Ontario's pipeline, by contrast, pivots toward clean electricity and mining.



*Figure 5: Heat Map of Closing Inventory by Province and Sector.*

Year-on-year trends reinforce these regional identities. Figure 6 clarifies how Alberta's inventory remains heavily concentrated in energy projects, while British Columbia, Ontario, and Québec

display more diversified sector portfolios. Mining shows momentum in British Columbia and Ontario, mirroring the national push for critical minerals, whereas forest-sector projects contract almost everywhere, save for a brief resurgence in Québec between 2019 and 2022, indicating limited new investment in forestry. Finally, pronounced year-to-year swings, such as Ontario's energy rebound and Québec's energy surge in 2024, demonstrate how a small number of large projects can quickly reshape provincial pipelines.



*Figure 6: Closing Inventory Trends by Sector Composition and Province.*

### *Project cost sensitivity*

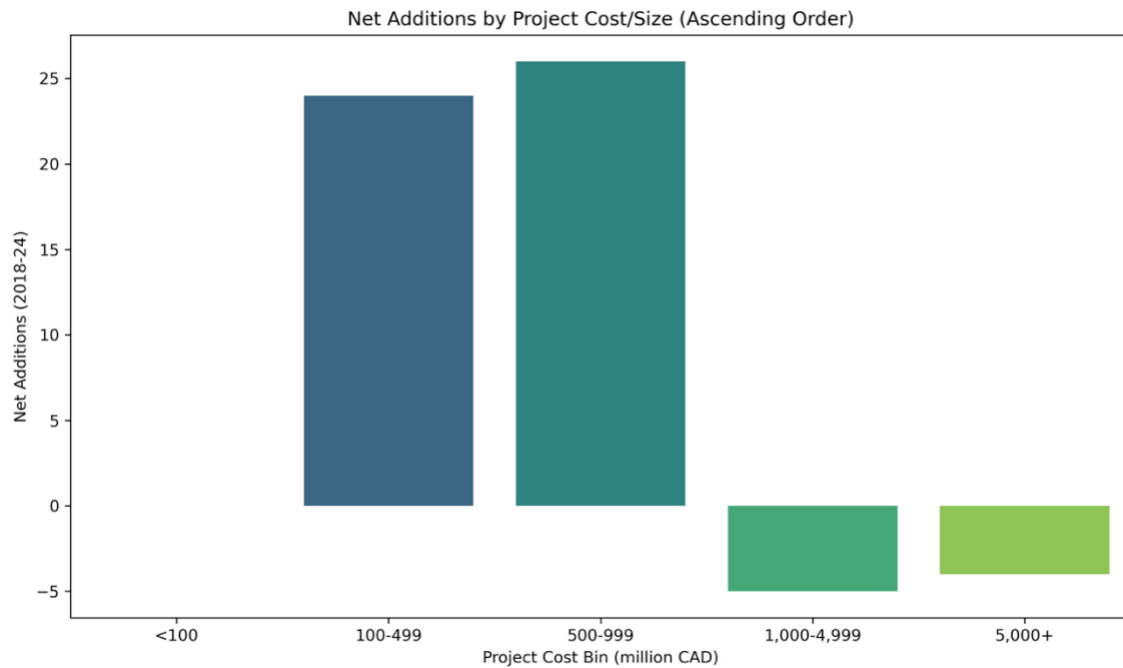
Table 2 summarizes project turnover from 2018 to 2024 by capital-cost band and shows that size matters. Small proposals costing under \$100 million and mid-tier developments between \$100 million and \$499 million generate the most activity, accounting for 1,172 of the

1,725 (68 percent) recorded entries and exits. By contrast, the headline-grabbing mega-projects, those budgeted at \$1 billion or more, shrink because departures outpace new filings.

| Cost Bin<br>C\$MM | Freshmen | Graduates | Dropouts | Net<br>Additions | Churn |
|-------------------|----------|-----------|----------|------------------|-------|
| <100              | 261      | 131       | 130      | 0                | 522   |
| 100-499           | 337      | 138       | 175      | 24               | 650   |
| 500-999           | 133      | 41        | 66       | 26               | 240   |
| 1,000-4,999       | 116      | 43        | 78       | -5               | 237   |
| 5,000+            | 36       | 12        | 28       | -4               | 76    |

*Table 2: Project Turnover by Project Size.*

Figure 7 visualizes net additions across the cost spectrum. All categories except for \$100 to \$999 million projects end the study period at or below their starting count.

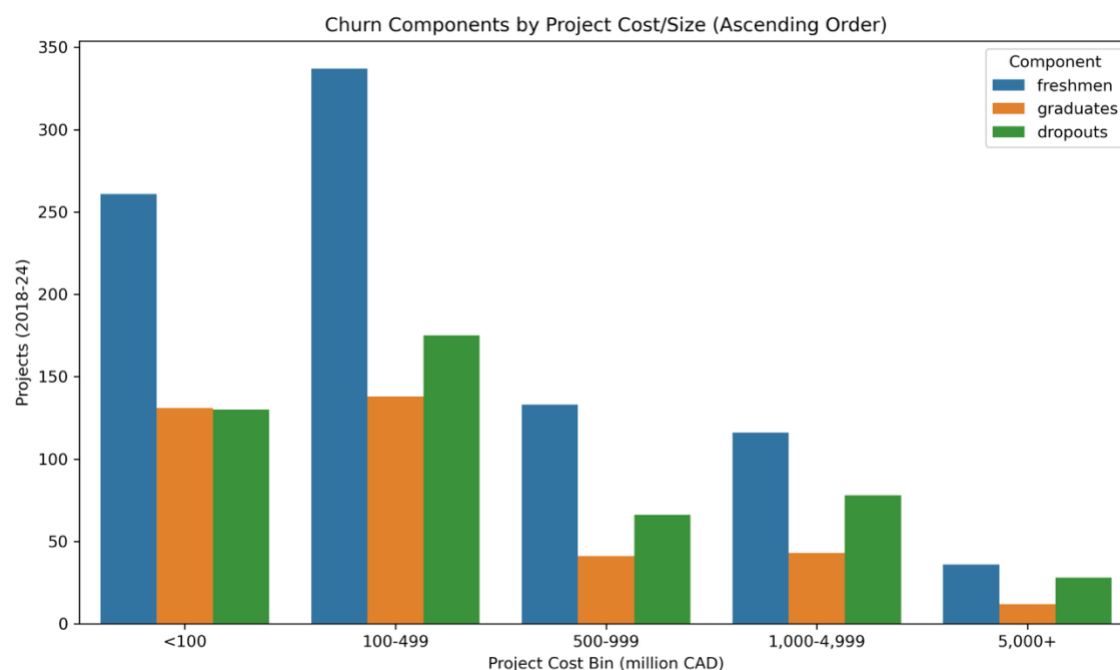


*Figure 7: Net Additions by Project Cost.*

The stacked columns in Figure 8 explain what drives these outcomes. Among \$1-billion-plus proposals, most exits are graduations, indicating that the projects are moving ahead, yet the trickle of fresh mega-projects is too small to offset the outflow. In the \$1 billion to \$4.9 billion

bracket, cancellations dominate, eroding the inventory despite some progress to construction.

Small projects behave differently. Every two departures are matched by roughly two new announcements, creating steady churn but leaving the overall tally essentially unchanged.



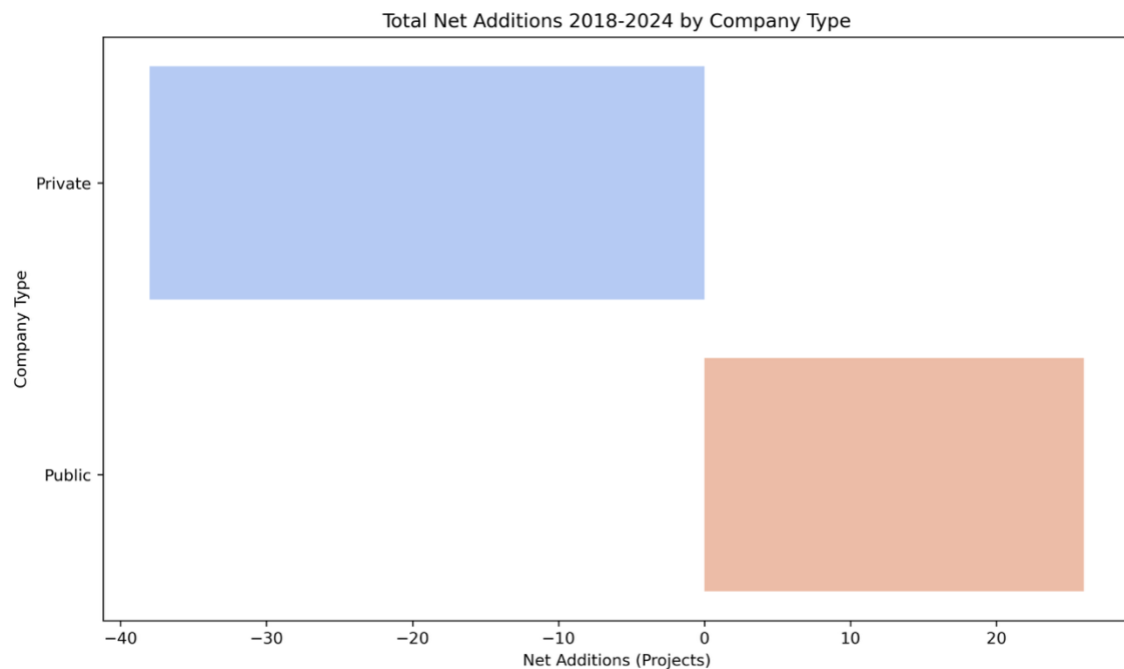
*Figure 8: Churn by Project Size and Stacked by Freshmen, Graduates, and Dropouts.*

### *Company ownership*

Company ownership shapes both the pace and risk profile of Canada's major-project pipeline. As Figure 9 shows, private-sector proponents drive nearly all activity, accounting for 1,382 project movements, about 85 percent of total churn, yet they finish the 2018-24 period with a modest net loss of 38 projects. The shortfall is largely explained by the sheer number of speculative, early-stage developments that private firms launch and later abandon. Public-sector sponsors, including crown corporations, municipalities and public utilities, appear far less often in the database, but when they do their projects tend to stick. Public entities add 26 net projects over

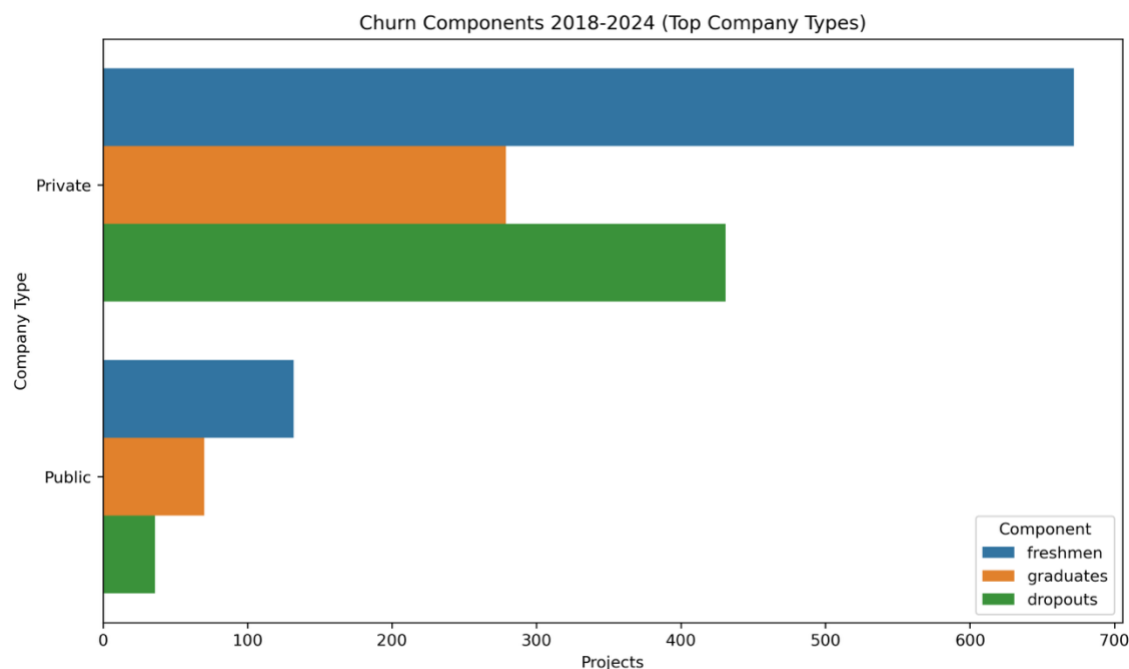
the same span, suggesting that once a public proposal enters the inventory it seldom disappears before reaching construction.

The contrast becomes even clearer in Figure 9. Bars extending to the left depict the private sector trimming its inventory, while those pointing right illustrate the public sector's incremental growth.



*Figure 9: Net Additions by Company Ownership*

Figure 10 shows private firms dominate both the freshman and dropout categories, suggesting a “fail-early” philosophy in which many concepts are tested and discarded quickly. Public sponsors, by contrast, register relatively few exits and display a higher proportion of graduates, a pattern consistent with more rigorous front-end vetting and stronger access to financing.



*Figure 10: Churn by Company Ownership*

Together, the figures underscore that public ownership confers a lower attrition rate and a higher likelihood of seeing projects through to construction, whereas the private sector remains the primary engine of fresh ideas, albeit with a higher propensity for early attrition.

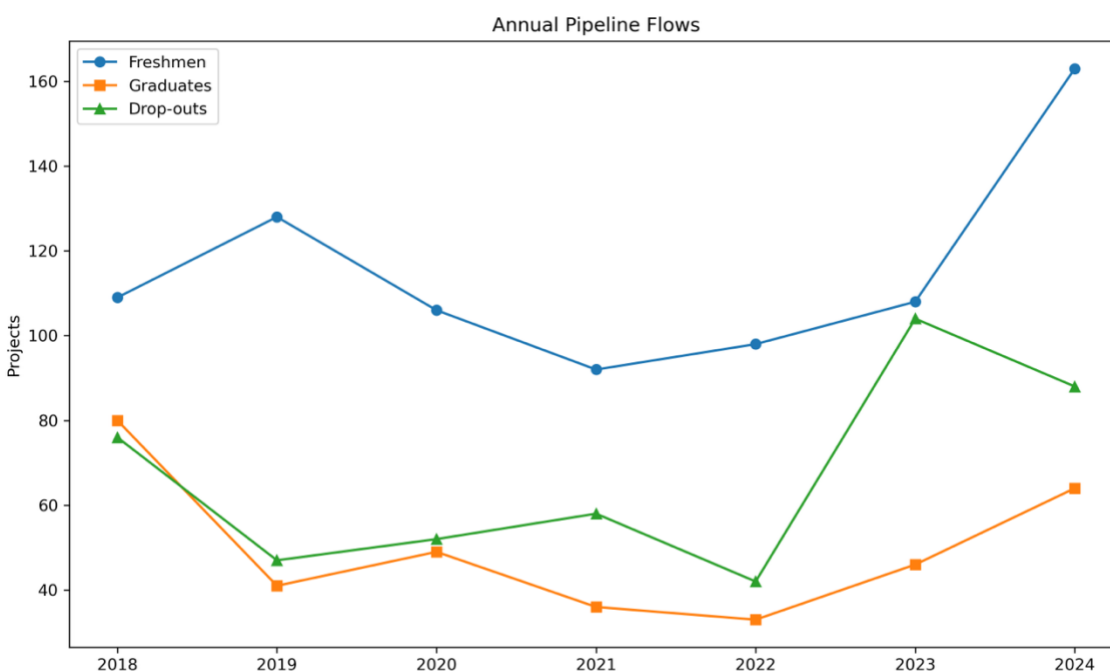
*Graduation bottleneck, freshmen and dropout spikes and macro shocks*

Annual flow of projects through the national pipeline reveals three distinct patterns. First, freshmen arrive in waves. The MPI database records a surge of 109 freshmen in 2018 and an even larger number of 163 in 2024. Second, graduations to “Under Construction” never exceed 80 projects in any year. Because completions lag filings, the graduation rate, calculated as the number of graduates divided by total exits, averages only about 45 percent across the period and plunges to 31 percent in 2023. Third, dropouts spike in 2023, when sponsors withdraw 104



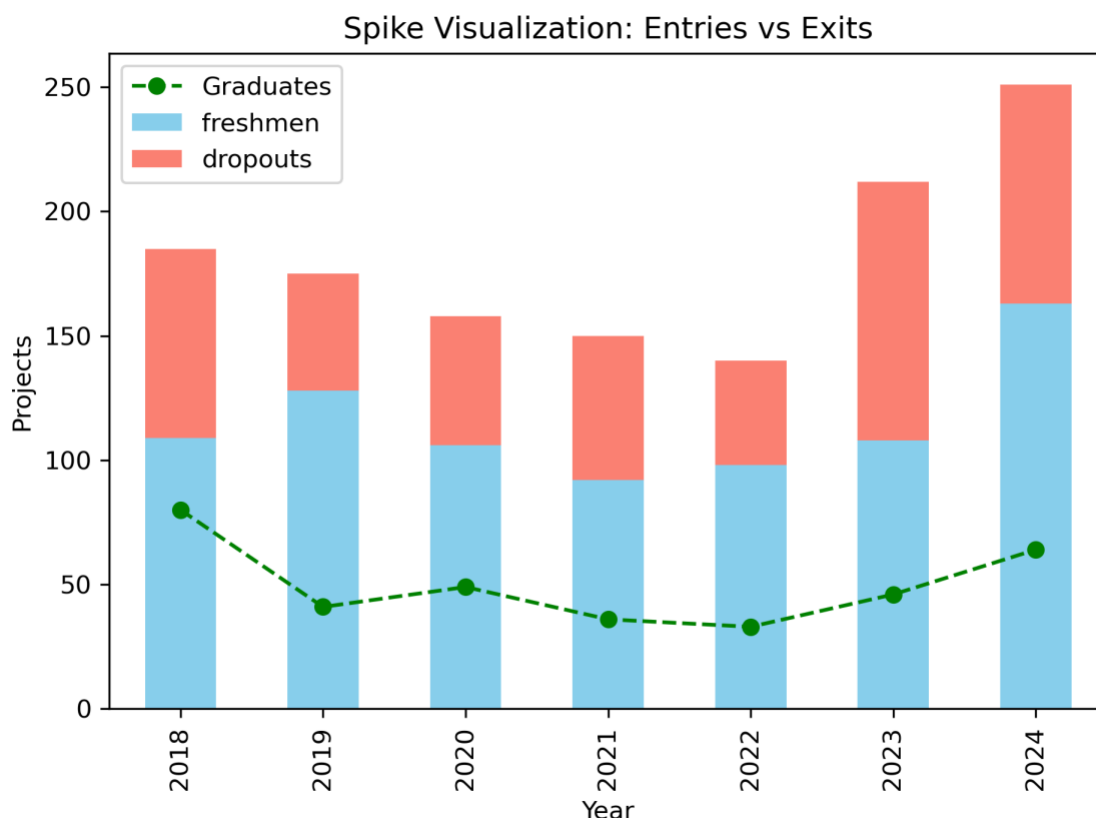
proposals, creating the widest gap on record between exits via cancellation and exits via completion.

The accompanying figures make the bottleneck visible. In Figure 11, the orange graduation line sits well below both the freshman and dropout curves every year except 2018, underscoring a persistent shortfall in projects that break ground.



*Figure 11: Annual Flow of Freshmen, Graduates and Dropouts by Year.*

Figure 12 expands the picture with stacked bars that separate the two main pressure points. In 2023 dropouts skyrocket to match freshmen, causing the overall inventory to contract. In 2024 a new wave of freshmen floods the queue just as graduations rebound, relieving the choke point yet pushing churn to an all-time high of 315 project movements.



*Figure 12: Entries vs. Exits of Freshmen, Graduates and Dropouts by Year.*

Taken together, the evidence points to a structural graduation bottleneck. Construction starts trail filing bursts by one to three years, so projects accumulate unless the system's capacity to graduate them expands in tandem. Cyclical investment windows, such as post-COVID stimulus, the critical-mineral rush, and clean policy initiatives, appear to inject large cohorts of freshmen. If a significant share later proves non-viable, they swell future dropout spikes. Conversely, when market conditions or policy signals deteriorate, marginal proposals retreat, freeing pipeline capacity but undermining near-term supply forecasts. In short, the health of Canada's major-project pipeline depends not only on the volume of new ideas (freshmen) but also on the system's ability to shepherd viable projects through to construction (graduates) before sentiment turns against them and they fall away (dropouts).

### *General discussion*

In Canada's major-project pipeline, freshmen rush in waves, graduates trickle out, and periodic dropout culls clear stalled files. Freshmen spikes in 2018 and 2024 coincide with policy stimulus and commodity up-cycles; yet only about half the projects that leave do so by advancing to construction, revealing a persistent graduation bottleneck. Regional and sectoral patterns sharpen the picture. Alberta and British Columbia record the highest churn, but the outcomes diverge. Alberta's heavy turnover erodes its pipeline, whereas British Columbia's boosts it. Québec, meanwhile, has quietly expanded its inventory by pairing steady inflows of new proposals with only moderate exits. On the sector side, hydrocarbons and grid projects continue to contract, while critical-mineral mining, clean electricity, and low-carbon fuels show modest growth.

The size analysis shows mid-tier (\$100 - \$999 million) projects filling the gap left by fewer multi-billion-dollar megaprojects, and small (<\$100 million) projects are feverishly treading water. Ownership splits confirm private proponents drive new project creation and volatility while public sponsors provide ballast and add durability. Altogether, the MPI appears both highly volatile and capacity constrained. Unless graduation rates rise, each new entry cohort will crowd the inventory and consume finite resources, heightening cancellation risk and delaying real-economy investment impacts.

The paper confirms a graduation bottleneck rooted in limited development capacity. Practical levers to address this bottleneck are already in motion. First, accelerated environmental reviews and consultation processes aim to expand graduation capacity. Second, the Canada Infrastructure Bank, Clean Fuels Fund, Canada Growth Fund and Investment Tax Credits channel billions of dollars of support toward C\$100 million to C\$1 billion projects, providing targeted incentives to

boost their completion record. Third, Canada's track record of pairing private developers with public sponsors can stabilize projects and reduce attrition.

Attention should now shift to emerging, often AI-enabled, tools that operate behind the scenes. Scenario models that vary commodity prices, government policy, and permitting timelines can stress-test pipeline capacity and flag impending commercial, financial and supply-chain bottlenecks. It is equally important to merge micro-level indicators, such as the sponsor's profile, permit milestones, financing details, offtake agreements, workforce metrics and material-cost indices, with MPI status codes so analysts can pinpoint whether a delay is primarily regulatory, financial, commercial or physical.

## **Conclusion**

The MPI depicts a restless pipeline of 500 projects, which struggles to convert proposals into operating assets. Each year roughly 230 projects enter or leave the inventory, yet fewer than half of the departures are true graduations to construction. Alberta and British Columbia dominate the volume of transitions, but their fortunes diverge. Alberta's heavy turnover erodes its pipeline, whereas British Columbia's inflows more than replace its exits. Québec, Manitoba and the Northwest Territories expand their inventories more quietly, adding modest numbers of projects with limited churn. Sector patterns are equally stark. Hydrocarbon and traditional grid assets continue to contract, while critical-mineral mining, clean electricity and emerging low-carbon fuels edge higher. Responsibility for net growth has shifted from multi-billion-dollar megaprojects to mid-sized ventures in the \$100 million to \$999 million range, which now account for most net additions. Volatility is sharpest among private proponents, who generate

most of both freshmen and drop-outs. Public sponsors, by contrast, retire far fewer projects prematurely and post consistent net gains.

These patterns imply a graduation bottleneck driven by finite development bandwidth, cyclical investment waves and the heightened risk of mega-projects. Relieving the constraint requires a twin approach. On the one hand, streamlined approvals, targeted federal financing (via CIB, CFF and CGF) and stronger public or Indigenous anchoring can raise near-term graduation capacity. On the other, data-driven surveillance and AI-enabled scenario modelling are needed to stress-test the pipeline under shifting conditions.

Future research could examine how macro- and micro-level factors influence project outcomes. Connecting data on financing, permitting, Indigenous agreements, and commercial contracts to MPI records could help reveal whether delays are primarily regulatory, financial, or physical. Cross-country comparisons might highlight the most promising regulatory or policy reforms. Ultimately, public anchoring and streamlined processes guided by data-driven planning and analysis offer a pragmatic path to turning Canada's steady flow of proposals into built assets.